DHCP — manzilni avtomatik berish

Dynamic Host Configuration Protocol

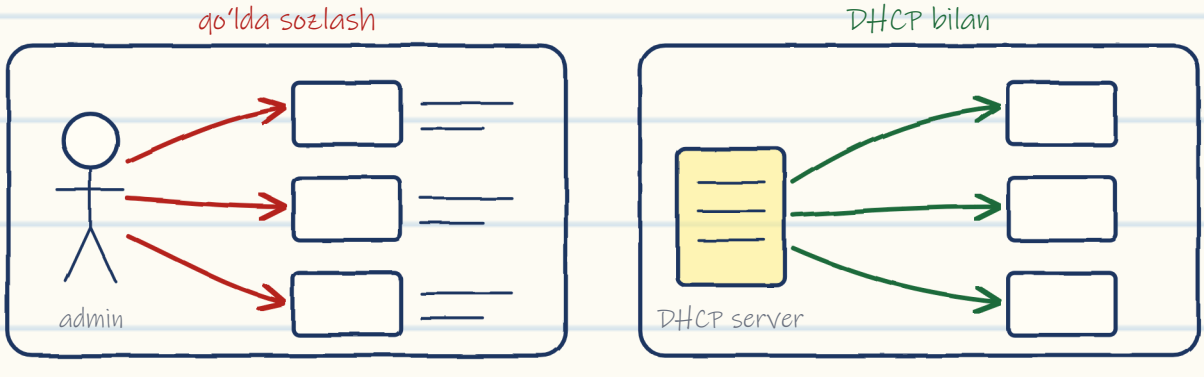
Asosiy atamalar

- DHCP — IP manzilni qurilmaga avtomatik beruvchi protokol
- Server va klient — manzil beruvchi va manzil so'rovchi tomon
- Pool (havza) — serverdagi tarqatish uchun ajratilgan manzillar to'plami
- Lease (ijara) — manzil berilgan muddat, tugasa yangilanadi
- DORA — 4 qadam: Discover → Offer → Request → ACK
- Relay agent — so'rovni boshqa tarmoqdagi serverga uzatuvchi qurilma

1. DHCP nima

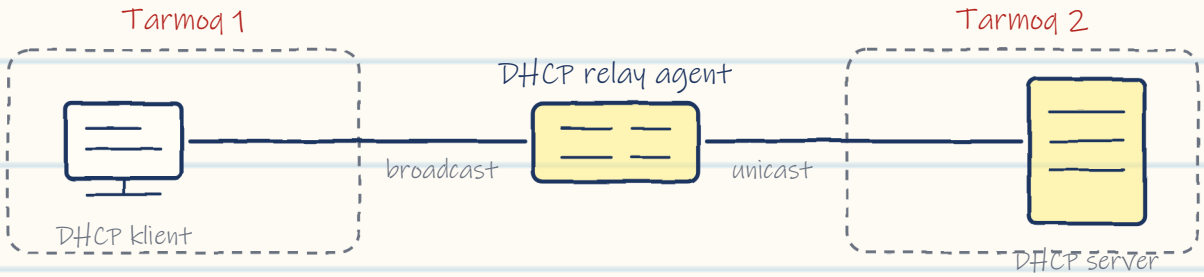
IP · maska · shlyuz · DNS — hammasini server avtomatik beradi.

- BOOTP ning kengaytmasi, u bilan mos
- UDP ustida: server 67 · klient 68
- klient/server — sozlamalar markazda turadi
- manzil ijaraga beriladi → resurs tejaladi



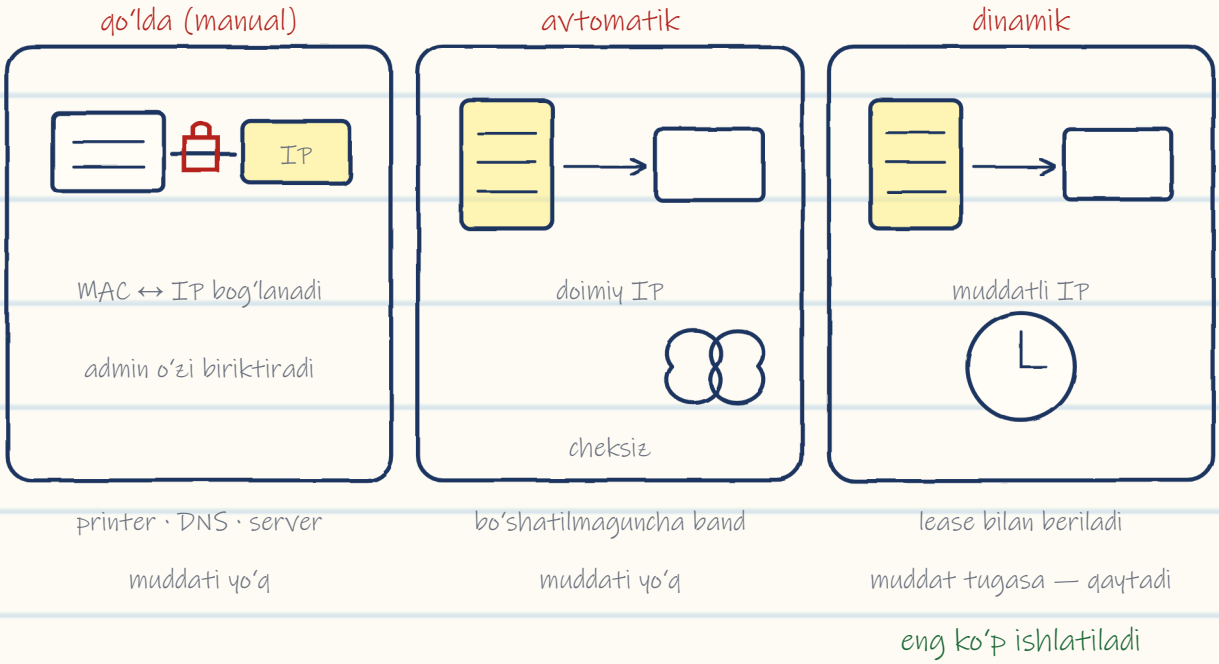
1-rasm. Katta tarmoqda qo'lda sozlash ish bermaydi

2. Tizim qismlari



2-rasm. Relay agent turli subnetlarni bog'laydi

3. Manzil taqsimlash usullari



3-rasm. Uch xil taqsimlash

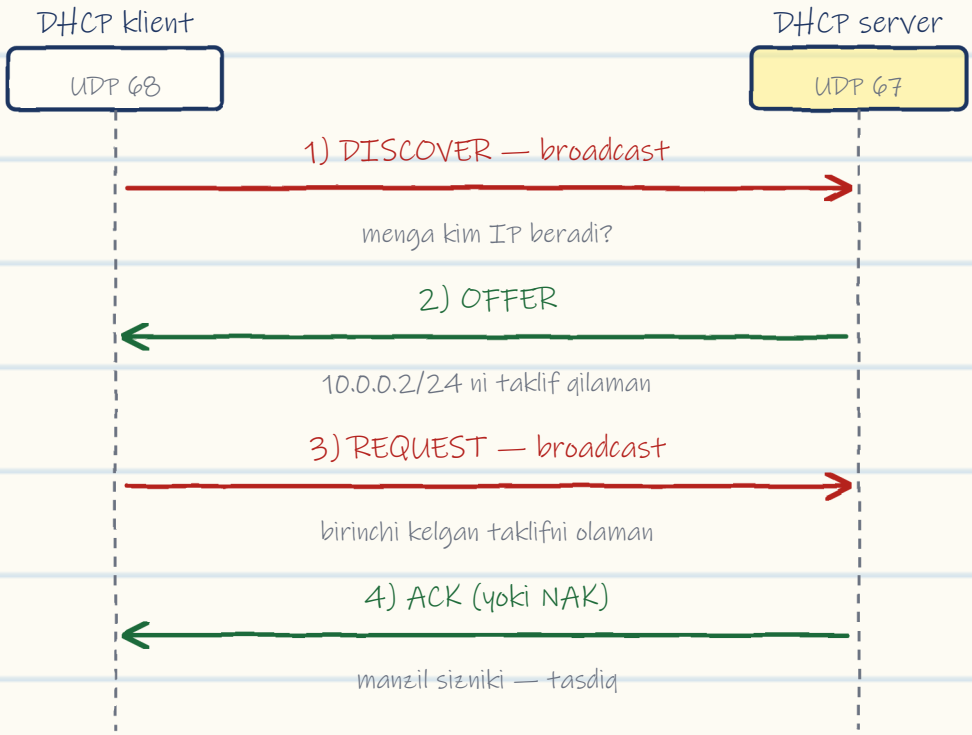
Usul	Muddat	Qayerda
Qo'lda	yo'q	printer, DNS, server
Avtomatik	doimiy	o'zgarmas xostlar
Dinamik	lease	oddiy foydalanuvchilar

4. DHCP nima beradi

- plug-and-play — xostni sozlash shart emas
- markazlashgan boshqaruv — barcha sozlama bitta joyda
- samaradorlik — ijara tugasa, manzil qayta ishlatiladi
- subnetlar aro — relay agent yordamida

5. Manzil olishning 4 bosqichi

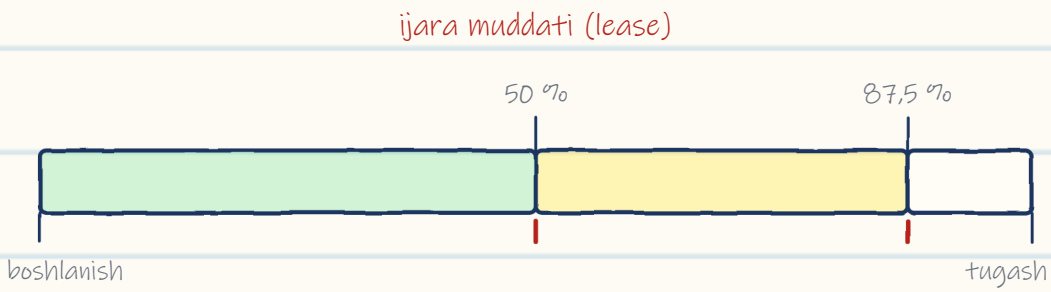
- DISCOVER — klient broadcast qiladi: server bormi?
- OFFER — server manzil taklif qiladi
- REQUEST — klient birinchi kelgan taklifni tanlaydi
- ACK — server tasdiqlaydi, NAK — rad etadi



4-rasm. DISCOVER → OFFER → REQUEST → ACK

Keyin: klient ARP bilan tekshiradi. Manzil band bo'lsa — DECLINE va yangi so'rov. Kerak bo'lmasa — RELEASE.

6. Ijara va uni yangilash



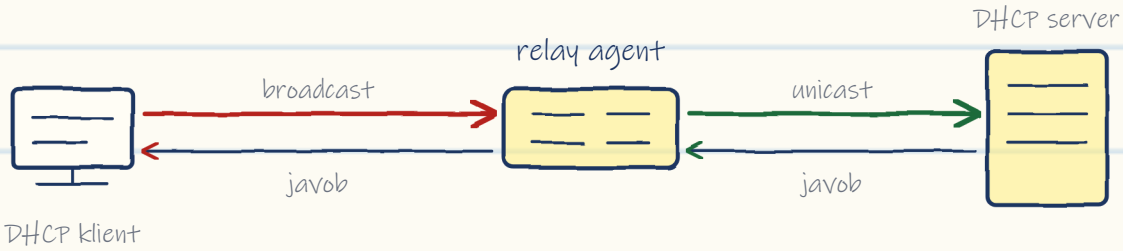
50 % — unicast REQUEST → o'sha serverga

87,5 % — broadcast REQUEST → istalgan serverga

5-rasm. Yangilash nuqtalari

Javob: ACK — ijara uzaytiriladi, NAK — rad, klient yangidan DISCOVER qiladi

7. Relay agent

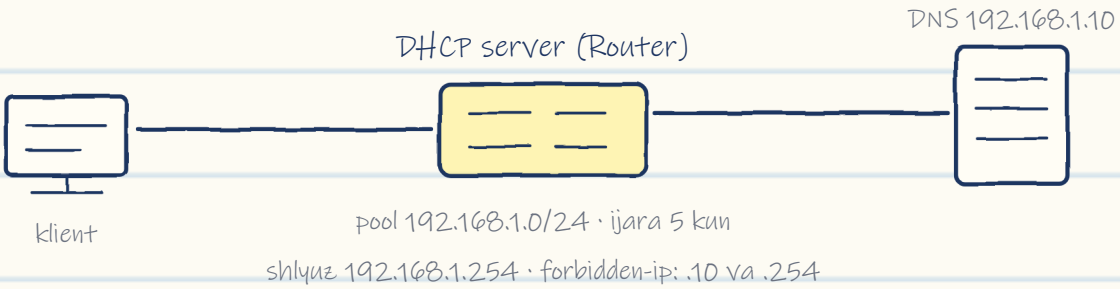


har subnetga alohida server kerak emas

6-rasm. Broadcast → unicast → server

- klientning broadcast paketini serverga unicast qilib uzatadi
- serverdan kelgan sozlamani klientga qaytaradi

8. Routerda sozlash



- [Router] dhcp enable
- [Router] dhcp server forbidden-ip 192.168.1.10
- [Router] dhcp server ip-pool 0
- [...pool-0] network 192.168.1.0 mask 255.255.255.0
- [...pool-0] gateway-list 192.168.1.254 · dns-list 192.168.1.10
- [...pool-0] expired day 5

Relay agent (interfeys ko'rinishida):

- dhcp select relay · dhcp relay server-address 192.168.1.10

Xulosa

- DHCP = BOOTP kengaytmasi · UDP 67 (server) / 68 (klient)
- Usullar: qo'lda · avtomatik · dinamik (lease)
- DISCOVER → OFFER → REQUEST → ACK
- Yangilash: 50 % unicast, 87,5 % broadcast
- Relay agent — server boshqa subnetda bo'lganda

Tekshirish

1. DHCP portlari? — 67 server, 68 klient
2. 4 bosqich? — DISCOVER, OFFER, REQUEST, ACK
3. Yangilash qachon? — 50 % va 87,5 %
4. Manzil band chiqsa? — DECLINE